

STAT 20000 Chapter 23

The Accuracy of Averages

A certain town has 50,000 people age 25 and over. As part of an opinion survey, a simple random sample of 400 such persons is taken. How to estimate the average year of schooling for all persons age 25 and over in this town?

- ▶ Population:
- ▶ Parameter — the **population average**:
 - the average year of schooling for persons in the population
- ▶ Statistic — the **sample average**:
 - the average year of schooling for persons in the sample

To know how close is the sample average from the population average, we need a box model.

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To know how close is the sample average from the population average, we need a box model.

A Box Model

| some 0's some 1's some 2's ... some 16's, ... |

- ▶ Each person in the population has 1 ticket in the box, showing _____.

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 - ▶ So there should be _____ tickets in the box.

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 - ▶ So there should be 50,000 tickets in the box.
- ▶ Sample average = $\frac{\text{Sum of draws}}{\text{Number of draws}}$.

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 - ▶ The sum of years of schooling in the sample is like the sum of _____ draws made at random _____ replacement from the box.

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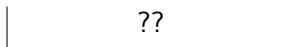
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- ▶ Sample average = $\frac{\text{Sum of draws}}{\text{Number of draws}}$.
 - ▶ The sum of years of schooling in the sample is like the sum of 400 draws made at random without replacement from the box.

Box Model for a SRS

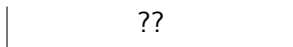
In general, when interested in the average of a certain quantities of subjects in some population, one can use the sample average of a SRS to estimate that. Properties of the sample average can be derived using a box model:



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- ▶ On the ticket shows

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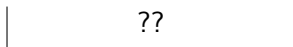
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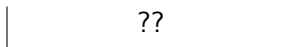
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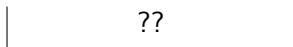
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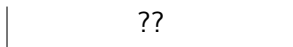
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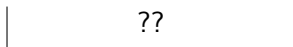
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- ▶ Parameter = Population average
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= Average of draws = $\frac{\text{Sum of draws}}{\text{Number of draws}}$.
- ▶ SD of the box = SD of the numbers in the box
(no shortcut formula since the tickets in the box usually have more than two distinct values.)

The Expected Value for the Sample Average

$$\text{Recall sample average} = \frac{\text{Sum of draws}}{\text{Number of draws}}.$$

When draws are made **with** replacement, the **expected value** for the **sum of draws** is

$$(\# \text{ of draws})(\text{Average of box})$$

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The **expected value** for the **sample average** is thus

$$\begin{aligned} & \frac{(\# \text{ of draws})(\text{Average of box})}{\# \text{ of draws}} \\ &= \text{Average of box} \\ &= \boxed{\text{population average}} \end{aligned}$$

The SE for the Sample Average

When draws are made **with** replacement, we know

$$\text{SE for the sum of draws} = \sqrt{\# \text{ of draws}} \times (\text{SD of box})$$

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A larger population does NOT require a larger sample!

Drawing Without Replacement v.s. With Replacement

The SE for the average of the draws when drawing *without* replacement is smaller than the corresponding SE for drawing *with* replacement by a factor of

$$\sqrt{\frac{\text{population size} - \text{sample size}}{\text{population size} - 1}}$$

- ▶ Same correction factor as used in Ch20&21
- ▶ Usually, the sample size is small compared to the population size, and the correction factor can be ignored.

Normal Approximation

For reasonably large samples, the **normal curve** can be used to figure out chances about sample averages, just as for the sum of the draws.

For example, the sample average is

- ▶ ≤ 1 SE from the population average w/ prob. about ... 68%;

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Warning: The facts listed above about sample averages apply to **simple random samples** only, but not to other kinds of sampling schemes.

Inferring the Population (Box) from the Sample (Draws)

In reality, the composition of the *box* is usually *unknown*, so are the *average and SD of the box*.

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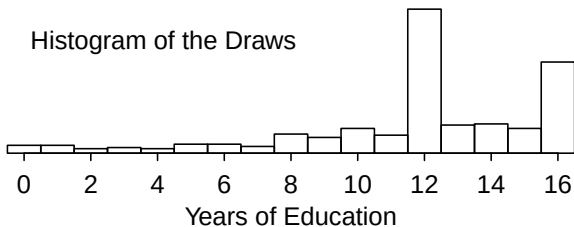
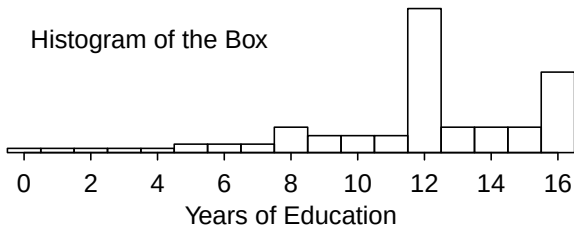
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- ▶ The unknown SD of box can be estimated by the SD of the sample — another **bootstrap method**.

Why is it okay to replace the SD of the box w/ the SD of the sample?



When the sample is large enough to be fairly representative of the whole box, the **SD of the sample** $\approx$ the **SD of the box**.

Confidence Intervals for Population Averages

Like population percentage, we can construct confidence intervals of various confidence levels for the population averages.

68%-C.I.	$\Rightarrow$	sample average $\pm$	1	SE
90%-C.I.	$\Rightarrow$	sample average $\pm$		SE
95%-C.I.	$\Rightarrow$	sample average $\pm$		SE
99%-C.I.	$\Rightarrow$	sample average $\pm$		SE

Here the SE is the estimated SE, where the SD of the box is estimated by the SD of the sample.

Confidence Intervals for Population Averages

Like population percentage, we can construct confidence intervals of various confidence levels for the population averages.

68%-C.I. $\Rightarrow$ sample average ± 1 SE

90%-C.I. $\Rightarrow$ sample average ± 1.64 SE

95%-C.I. $\Rightarrow$ sample average ± 1.96 SE

99%-C.I. $\Rightarrow$ sample average ± 2.58 SE

Here the SE is the estimated SE, where the SD of the box is estimated by the SD of the sample.

The SE for the average of 400 draws from the box is estimated as

$$\frac{\text{SD of sample}}{\sqrt{\text{sample size}}} = \frac{4.1 \text{ years}}{\sqrt{400}} \approx 0.2 \text{ years.}$$

The average educational level in the town can be estimated as _____ years, give or take _____ years.

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A 95%-confidence interval for the average year of schooling in the town is

$$\text{sample average} \pm 2 (\text{estimated SE}) = 11.6 \pm 0.4 \text{ years.}$$

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True or false, and explain: There is a 95%-chance that the population average lies between 11.2 years and 12.0 years.

What's the thing that has a 95% chance to happen?

Which SE?

We have formulas for the SEs for several chance processes, all of which involve drawing at random w/ replacement from a box:

SE for the	Formula
Sum of draws	
Ave. of draws	
Count	
Percent	

Here $n = \#$ of draws, and $p =$ the fraction of 1's in the box.

When the box (the population) is unknown, you have to estimate the SD of the box from the sample.

- ▶ For 0–1 boxes, estimate p by the fraction of 1's in the sample
- ▶ For other boxes, estimate the SD of the box by the SD of the sample. (Note: The $\#$ of draws needs to be reasonably large.)

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Sum of draws	$(\text{SD of box}) \times \sqrt{n}$
Ave. of draws	$(\text{SD of box}) / \sqrt{n}$
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Percent	$\sqrt{p(1-p)} / \sqrt{n} \times 100\%$

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Example: Utility Company Survey I

A utility company serves 50,000 households. As a part of a survey of customer attitudes, they take a SRS of 400 of these households. The average number of TV sets in the sample households turns out to be 1.86, and the SD is 0.90.

If possible, find a 95%-confidence interval for the average number of TV sets in all 50,000 households. If this isn't possible, explain why.

- ▶ This is _____. The box model has 1 ticket for each household, showing the _____.
- ▶ Estimate the average of box as _____.
- ▶ Estimate the SD of box as _____.
- ▶ Estimate the SE of average of draws as _____.

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- ▶ Estimate the SD of box as 0.90.
- ▶ Estimate the SE of average of draws as _____.

$$\frac{\text{SD of sample}}{\sqrt{\text{sample size}}} = \frac{0.90}{\sqrt{400}} = 0.045$$

Example: Utility Company Survey I

A utility company serves 50,000 households. As a part of a survey of customer attitudes, they take a SRS of 400 of these households. The average number of TV sets in the sample households turns out to be 1.86, and the SD is 0.90.

If possible, find a 95%-confidence interval for the average number of TV sets in all 50,000 households. If this isn't possible, explain why.

- ▶ This is possible. The box model has 1 ticket for each household, showing the # of TV sets in the household.
- ▶ Estimate the average of box as 1.86.
- ▶ Estimate the SD of box as 0.90.
- ▶ Estimate the SE of average of draws as 0.045.

$$\frac{\text{SD of sample}}{\sqrt{\text{sample size}}} = \frac{0.90}{\sqrt{400}} = 0.045$$

Example: Utility Company Survey I (Cont'd)

- ▶ 95% confidence interval is _____.

That is, with 95% confidence, we may assert that the average number of television sets in all 50,000 households is somewhere between _____ and _____.

Example: Utility Company Survey I (Cont'd)

- ▶ 95% confidence interval is _____.

$$\begin{aligned} & \text{estimate} \pm 2 \times (\text{SE of estimate}) \\ & = 1.86 \pm 2 \times 0.045 = 1.86 \pm 0.09. \end{aligned}$$

That is, with 95% confidence, we may assert that the average number of television sets in all 50,000 households is somewhere between _____ and _____.

Example: Utility Company Survey I (Cont'd)

- 95% confidence interval is _____.

$$\begin{aligned} & \text{estimate} \pm 2 \times (\text{SE of estimate}) \\ &= 1.86 \pm 2 \times 0.045 = 1.86 \pm 0.09. \end{aligned}$$

That is, with 95% confidence, we may assert that the average number of television sets in all 50,000 households is somewhere between $1.86 - 0.09 = 1.77$ and $1.86 + 0.09 = 1.95$.

Example: Utility Company Survey I (Cont'd)

- ▶ 95% confidence interval is 1.77 to 1.95.

$$\begin{aligned} & \text{estimate} \pm 2 \times (\text{SE of estimate}) \\ &= 1.86 \pm 2 \times 0.045 = 1.86 \pm 0.09. \end{aligned}$$

That is, with 95% confidence, we may assert that the average number of television sets in all 50,000 households is somewhere between $1.86 - 0.09 = 1.77$ and $1.86 + 0.09 = 1.95$.

Example: Utility Company Survey II

Out of the 400 households in the survey, 187 had Netflix.

If possible, find an approximate 90% confidence interval for the percentage of all the 50,000 households having Netflix. If this isn't possible, explain why.

- ▶ This is _____.
The box model has one ticket for each household, showing _____ if having Netflix, and _____ otherwise.
- ▶ The population percentage is estimated as _____
- ▶ The SD of box is estimated as _____

Example: Utility Company Survey II

Out of the 400 households in the survey, 187 had Netflix.

If possible, find an approximate 90% confidence interval for the percentage of all the 50,000 households having Netflix. If this isn't possible, explain why.

- ▶ This is possible.
The box model has one ticket for each household, showing _____ if having Netflix, and _____ otherwise.
- ▶ The population percentage is estimated as
- ▶ The SD of box is estimated as

Example: Utility Company Survey II

Out of the 400 households in the survey, 187 had Netflix.

If possible, find an approximate 90% confidence interval for the percentage of all the 50,000 households having Netflix. If this isn't possible, explain why.

- ▶ This is possible.
The box model has one ticket for each household, showing 1 if having Netflix, and 0 otherwise.
- ▶ The population percentage is estimated as
- ▶ The SD of box is estimated as

Example: Utility Company Survey II

Out of the 400 households in the survey, 187 had Netflix.

If possible, find an approximate 90% confidence interval for the percentage of all the 50,000 households having Netflix. If this isn't possible, explain why.

- ▶ This is possible.

The box model has one ticket for each household, showing 1 if having Netflix, and 0 otherwise.

- ▶ The population percentage is estimated as

$$\frac{187}{400} \times 100\% = 46.75\%.$$

- ▶ The SD of box is estimated as

Example: Utility Company Survey II

Out of the 400 households in the survey, 187 had Netflix.

If possible, find an approximate 90% confidence interval for the percentage of all the 50,000 households having Netflix. If this isn't possible, explain why.

- ▶ This is possible.

The box model has one ticket for each household, showing 1 if having Netflix, and 0 otherwise.

- ▶ The population percentage is estimated as

$$\frac{187}{400} \times 100\% = 46.75\%.$$

- ▶ The SD of box is estimated as

$$\begin{aligned} & \sqrt{(\text{fraction of 1's in sample}) \times (\text{fraction of 0's in sample})} \\ &= \sqrt{0.4675 \times (1 - 0.4675)} = 0.499 \approx 0.5 \end{aligned}$$

Example: Utility Company Survey II (Cont'd)

- ▶ Estimate the SE of the sample percentage as _____.
- ▶ 90% confidence interval is _____.

That is, with 90% confidence, we may assert that the percentage of all the 50,000 households having Netflix is somewhere between _____ and _____.

Example: Utility Company Survey II (Cont'd)

- ▶ Estimate the SE of the sample percentage as _____.

$$\frac{\text{SD of sample}}{\sqrt{\text{sample size}}} \times 100\% = \frac{0.5}{\sqrt{400}} \times 100\% = 2.50\%.$$

- ▶ 90% confidence interval is _____.

That is, with 90% confidence, we may assert that the percentage of all the 50,000 households having Netflix is somewhere between _____ and _____.

Example: Utility Company Survey II (Cont'd)

- ▶ Estimate the SE of the sample percentage as 2.50%.

$$\frac{\text{SD of sample}}{\sqrt{\text{sample size}}} \times 100\% = \frac{0.5}{\sqrt{400}} \times 100\% = 2.50\%.$$

- ▶ 90% confidence interval is _____.

That is, with 90% confidence, we may assert that the percentage of all the 50,000 households having Netflix is somewhere between _____ and _____.

Example: Utility Company Survey II (Cont'd)

- ▶ Estimate the SE of the sample percentage as 2.50%.

$$\frac{\text{SD of sample}}{\sqrt{\text{sample size}}} \times 100\% = \frac{0.5}{\sqrt{400}} \times 100\% = 2.50\%.$$

- ▶ 90% confidence interval is _____.

$$\begin{aligned} & \text{estimate} \pm 1.64 \times (\text{SE of estimate}) \\ & = 46.75\% \pm 1.64 \times 2.50\% = 46.75\% \pm 4.10\% \end{aligned}$$

That is, with 90% confidence, we may assert that the percentage of all the 50,000 households having Netflix is somewhere between _____ and _____.

Example: Utility Company Survey II (Cont'd)

- ▶ Estimate the SE of the sample percentage as 2.50%.

$$\frac{\text{SD of sample}}{\sqrt{\text{sample size}}} \times 100\% = \frac{0.5}{\sqrt{400}} \times 100\% = 2.50\%.$$

- ▶ 90% confidence interval is _____.

$$\begin{aligned} & \text{estimate} \pm 1.64 \times (\text{SE of estimate}) \\ & = 46.75\% \pm 1.64 \times 2.50\% = 46.75\% \pm 4.10\% \end{aligned}$$

That is, with 90% confidence, we may assert that the percentage of all the 50,000 households having Netflix is somewhere between 46.75% - 4.10% = 42.65% and 46.75% + 4.10% = 50.85%.

Example: Utility Company Survey II (Cont'd)

- ▶ Estimate the SE of the sample percentage as 2.50%.

$$\frac{\text{SD of sample}}{\sqrt{\text{sample size}}} \times 100\% = \frac{0.5}{\sqrt{400}} \times 100\% = 2.50\%.$$

- ▶ 90% confidence interval is 42.63% to 50.87%.

$$\begin{aligned} & \text{estimate} \pm 1.64 \times (\text{SE of estimate}) \\ & = 46.75\% \pm 1.64 \times 2.50\% = 46.75\% \pm 4.10\% \end{aligned}$$

That is, with 90% confidence, we may assert that the percentage of all the 50,000 households having Netflix is somewhere between $46.75\% - 4.10\% = 42.65\%$ and $46.75\% + 4.10\% = 50.85\%$.

Example: Utility Company Survey III

As part of the survey, all persons age 16 and over in the 400 sample households are interviewed. This makes 900 people.

On the average, the sample people watched 5.20 hours of TV the Sunday before the sample, and the SD was 4.50 hours.

If possible, find a 95%-confidence interval for the average number of hours spent watching TV by all persons age 16 and over in the 50,000 households on that Sunday. If this isn't possible, explain why.

► This is _____, because

► Our formulas don't apply.

Example: Utility Company Survey III

As part of the survey, all persons age 16 and over in the 400 sample households are interviewed. This makes 900 people.

On the average, the sample people watched 5.20 hours of TV the Sunday before the sample, and the SD was 4.50 hours.

If possible, find a 95%-confidence interval for the average number of hours spent watching TV by all persons age 16 and over in the 50,000 households on that Sunday. If this isn't possible, explain why.

► This is not possible, because

► Our formulas don't apply.

Example: Utility Company Survey III

As part of the survey, all persons age 16 and over in the 400 sample households are interviewed. This makes 900 people.

On the average, the sample people watched 5.20 hours of TV the Sunday before the sample, and the SD was 4.50 hours.

If possible, find a 95%-confidence interval for the average number of hours spent watching TV by all persons age 16 and over in the 50,000 households on that Sunday. If this isn't possible, explain why.

- ▶ This is not possible, because these people are NOT a **SRS** from the target population. There will be dependencies among the hours of TV watched among members of the same household.
- ▶ Our formulas don't apply.

Summary

- ▶ When drawing at random from a box, the expected value for the average of the draws equals the average of the box. And the SE for the average of the draws equals the SE for their sum, divided by the number of draws. For sampling with replacement, the SE for the average of the draws works out to $(\text{SD of the box})/\sqrt{\text{number of draws}}$.
- ▶ The sample average can be used to estimate the population average. The estimate will be off by some amount, due to chance error. The SE for the sample average of the draws tells you the likely size of the amount off.
- ▶ Multiplying the number of draws by some factor divides the SE for their average by the square root of that factor.

Summary (Cont'd)

- ▶ The probability histogram for the average of the draws will follow the normal curve, even if the contents of the box do not. The histogram must be put into standard units, and the number of draws must be reasonably large.
- ▶ With a simple random sample, the SD of the sample can be used to estimate the SD of the box. A confidence interval for the average of the box can be found by going the right number of SEs either way from the average of the draws. The confidence level is read off the normal curve. This method should only be used with large samples.
- ▶ The formulas for simple random samples should not be applied mechanically to other kinds of samples.